Data
Analytics

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Sales Performance

A production analytics dashboard built
entirely inside Google Sheets.

ROLE: Design + Development

STACK: Apps Script · Chart.js · Tailwind

TIMELINE: 4 Days

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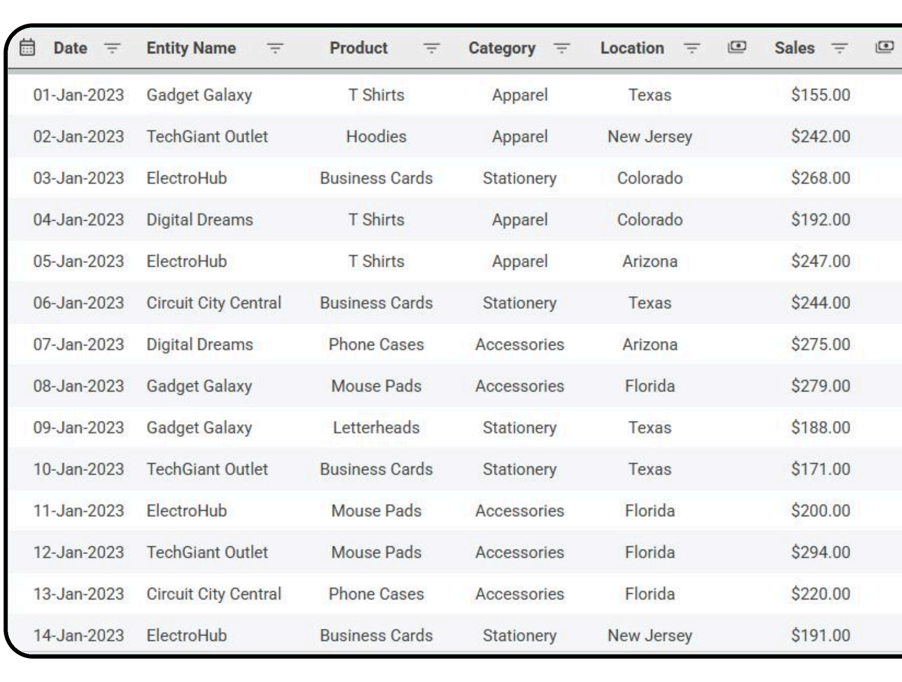
Problem

A production analytics dashboard built entirely inside Google Sheets.



Date	Entity Name	Product	Category	Location	Sales
01-Jan-2023	Gadget Galaxy	T Shirts	Apparel	Texas	$155.00
02-Jan-2023	TechGiant Outlet	Hoodies	Apparel	New Jersey	$242.00
03-Jan-2023	ElectroHub	Business Cards	Stationery	Colorado	$268.00
04-Jan-2023	Digital Dreams	T Shirts	Apparel	Colorado	$192.00
05-Jan-2023	ElectroHub	T Shirts	Apparel	Arizona	$247.00
06-Jan-2023	Circuit City Central	Business Cards	Stationery	Texas	$244.00
07-Jan-2023	Digital Dreams	Phone Cases	Accessories	Arizona	$275.00
08-Jan-2023	Gadget Galaxy	Mouse Pads	Accessories	Florida	$279.00
09-Jan-2023	Gadget Galaxy	Letterheads	Stationery	Texas	$188.00
10-Jan-2023	TechGiant Outlet	Business Cards	Stationery	Texas	$171.00
11-Jan-2023	ElectroHub	Mouse Pads	Accessories	Florida	$200.00
12-Jan-2023	TechGiant Outlet	Mouse Pads	Accessories	Florida	$294.00
13-Jan-2023	Circuit City Central	Phone Cases	Accessories	Florida	$220.00
14-Jan-2023	ElectroHub	Business Cards	Stationery	New Jersey	$191.00

I took a raw sales dataset of 12 columns covering transactions across entities, products, categories, and locations then asked what it would take to actually read it. Every column needed is already there. Reading it is the part that doesn't scale.

A spreadsheet holds the data. It doesn't hold the answers.

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Approach

Build the analytics layer
where the data already lives.

The obvious path was to export the data and rebuild it somewhere better. I went the other way. If the data lives in a sheet, the dashboard should too. That single constraint decided everything that followed.

How the data moves

Google Sheet	Apps Script	HTML service	Browser
731 rows	aggregation	Chart.js	filters

SERVER SIDE — CLIENT SIDE

Three decisions that shaped the build

AGGREGATE ON THE SERVER
731 transactions leave the sheet as roughly 20 numbers. The browser receives answers, not rows.

MATCH COLUMNS BY NAME
Headers are read at runtime, not hardcoded by position. Reorder a column and nothing breaks.

FILL THE GAPS
Empty months are zeros, not missing rows. Otherwise the prior period lines up wrong.

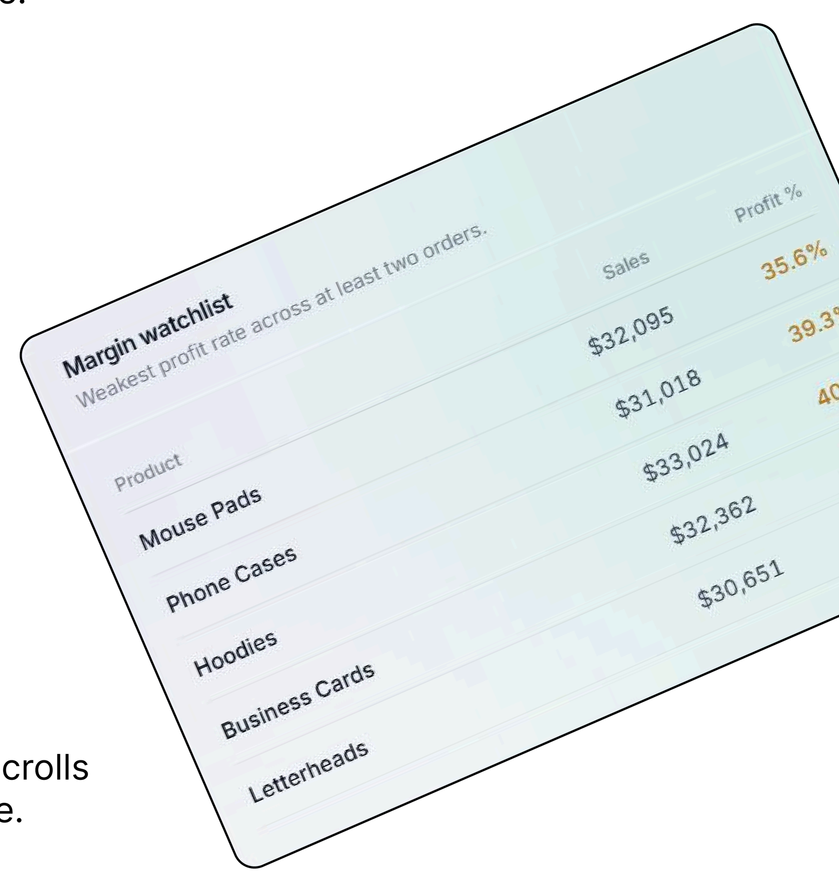
03 /

Design System

A system, not a screen.

One typeface. Four accents. Colour reserved for the two things that carry meaning: profit and loss.

Colour here does one of two jobs. It decorates, or it carries meaning. Doing the second rules out the first. Green means profit everywhere it appears. Four accents carry meaning, four neutrals carry everything else.



A dashboard is columns of figures, and in most typefaces a 1 is narrower than an 8, so columns drift. Inter's tabular numerals give every digit the same width. One typeface handles headings and figures, so the interface speaks in a single voice.

Glass came last, and only because it earned a function. The filter bar stays fixed while content scrolls beneath it. Blur is what makes that layering legible.

04 /

Analytics Logic

Every chart answers a question a table couldn't.

Charts are easy to add and hard to justify. Each one here replaced a question that the raw sheet could not answer without a pivot table built by hand.

/WATERFALL
Where does the money go?
Sales, cost, margin, expenses, profit as one falling sequence. A KPI grid shows the same five numbers but hides the relationship between them.

/PARETO
How concentrated is revenue?
Bars ranked by sales with a cumulative line above. A top-ten list shows the winners. This shows whether three products carry the business or thirty do.

/PRIOR PERIOD OVERLAY
Is this normal?
A dotted line of the preceding window, month for month. Without it every figure is a snapshot with nothing to compare against.

/MARGIN BUBBLE
Which category should grow?
Revenue against margin rate, sized by profit. Sorting by revenue alone rewards the loudest category, not the most profitable one.

The dashboard has no chart that only restates a number already shown in the KPI rail.

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What Broke

Every chart answers a question a table couldn't.

Challenge 1

MONTHS THAT WENT MISSING

The prior-period line compared the wrong months and the chart looked correct while doing it.

The series was built only from months that had rows. Filter to a location with a quiet March and that month vanished from the array. Twelve current months lined up against ten prior ones by index, so January was compared to March.

The fix was to generate every month between the two dates first, then fill. A month with no sales is a zero, not a missing entry.

Challenge 2

THE CACHE THAT DIDN'T FIT

CacheService caps a single entry at 100KB, so the write is guarded at 95,000 bytes. I assumed that was headroom.

Measured:
731 rows
138,255 bytes
189 bytes per row
limit 95,000

Every call had been reading the sheet live. The guard skipped the write silently, and nothing in the logs said so.

One produced wrong numbers that looked right. The other produced right numbers slower than it claimed. Neither showed up on screen.

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Outcome

Questions that took a pivot table now take a click.

Before	After
Which market is losing margin?	
was: build a pivot, sort, cross-check	now: one bar chart, sorted by profit rate
Is this month normal?	
was: no way to tell from the sheet	now: prior period drawn beneath every bar
Where does revenue actually come from?	
was: sort by sales, guess at the tail	now: cumulative line shows concentration
What is quietly unprofitable?	
was: never surfaced	now: margin watchlist, ranked

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